

PRODUCT REQUIREMENTS DOCUMENT

Industrial Knowledge Intelligence Platform

An AI Knowledge Layer for Industrial Organizations

Personal PRD · Version 0.1

Working Document — Alignment Draft, Not Final

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Table of Contents

Table of Contents 2

1. What Are We Building?..... 3

2. What Problem Are We Solving? 3

3. Product Vision 4

4. Success Criteria..... 4

5. Who Are Our Users? 4

6. Core Product Capabilities 5

7. User Flow..... 8

8. Product Boundaries 8

9. MVP Scope (10–15 Days) 9

10. Guiding Principles 10

1. What Are We Building?

We're building an AI-powered Industrial Knowledge Intelligence Platform.

The platform helps industrial organizations centralize scattered engineering knowledge, understand it using AI, connect related information across different document types, and make that knowledge instantly searchable and actionable.

Instead of engineers manually searching across folders, emails, maintenance systems, SOPs, and PDFs, the platform becomes a single place where organizational knowledge can be queried and continuously updated.

The product is not another document management system.

The product is an AI knowledge layer built on top of industrial documents.

2. What Problem Are We Solving?

Industrial companies already have documentation.

The problem is that the knowledge is fragmented.

Example

- Maintenance history is stored in one system.
- Engineering drawings are somewhere else.
- Inspection reports exist as PDFs.
- SOPs are stored in SharePoint.
- Emails contain critical decisions.
- OEM manuals sit in folders.

When an engineer needs information, they spend hours searching.

Sometimes they never find it.

Sometimes the same issue has already happened years earlier, but nobody knows.

The company loses knowledge every time experienced employees retire.

3. Product Vision

Create one AI system that continuously learns from every industrial document inside an organization and becomes the company's operational memory.

It should answer questions, connect related knowledge, help maintenance teams, assist compliance teams, and reduce time spent searching for information.

4. Success Criteria

If the platform is successful, users should be able to:

- Upload any industrial document.
- Search across every document from one place.
- Ask natural language questions.
- Receive grounded answers with citations.
- Discover related documents automatically.
- Find previous incidents for the same equipment.
- Access knowledge from mobile or desktop.
- Keep organizational knowledge even after employees leave.

5. Who Are Our Users?

Persona	Core Needs
Plant Engineer	Engineering drawings, manuals, historical decisions, equipment information.
Maintenance Engineer	Previous failures, work orders, inspection history, replacement procedures.
Field Technician	SOPs, maintenance instructions, safety procedures — from a phone.
Compliance Officer	Regulations, inspections, audit evidence.
Plant Manager	Visibility across documents and operational knowledge.

6. Core Product Capabilities

Instead of thinking about features, think about capabilities.

Capability 1 — Universal Document Ingestion

The system should accept:

- PDF
- DOCX
- Excel
- CSV
- Images
- Scanned documents
- Emails
- Engineering drawings
- P&IDs

Users shouldn't care about file type.

Capability 2 — Document Understanding

Uploading isn't enough. The system must understand documents. It should:

- OCR scanned pages
- Detect tables
- Identify engineering drawings
- Extract metadata
- Understand document structure

Capability 3 — Knowledge Extraction

After understanding a document, the system should identify:

- Equipment
- People
- Locations
- Regulations
- Maintenance actions
- Failures
- Inspection dates

- Operating parameters

These become structured knowledge.

Capability 4 — Knowledge Connections

Knowledge shouldn't remain isolated. The system should automatically connect related records — for example:

- **Pump P101**
- **Maintenance Record**
- **OEM Manual**
- **Inspection Report**
- **Failure Report**
- **Replacement Procedure**

Everything should become connected.

Capability 5 — Enterprise Search

Users should search using:

- Keywords
- Equipment names
- Natural language
- Document filters
- Semantic search

The platform should search knowledge, not filenames.

Capability 6 — AI Knowledge Assistant

Users should be able to ask, for example: "What caused Pump P101 to fail last year?"

The assistant should:

- Search
- Retrieve
- Reason
- Answer
- Provide citations
- Link documents

- Show confidence

Capability 7 — Knowledge Discovery

Users shouldn't only get answers. The system should proactively show:

- Similar incidents
- Related equipment
- Relevant manuals
- Previous maintenance
- Similar failures

Capability 8 — Continuous Learning

Whenever a new document is uploaded, the knowledge base should automatically update itself.

Nobody should rebuild indexes manually.

7. User Flow

A normal workflow looks like this:

- Upload document
- Document stored
- Document processed
- Knowledge extracted
- Knowledge connected
- Search index updated
- AI can use it
- User asks question
- Answer generated
- Source documents shown

8. Product Boundaries

We ARE Building	We Are NOT Building
AI-powered document intelligence	ERP
Enterprise search	SAP replacement
Knowledge extraction	CMMS replacement
AI assistant	CAD editor
Knowledge relationships	Email client
Industrial document management	Asset monitoring system / IoT platform / Real-time sensor analytics

Those systems may integrate later, but they aren't our responsibility.

9. MVP Scope (10–15 Days)

Must Have

- User login
- Document upload
- OCR / parsing
- Chunking
- Embeddings
- Vector search
- AI chat
- Source citations
- Document viewer
- Basic metadata extraction

Should Have

- Equipment entity extraction
- Relationship extraction
- Knowledge graph (backend)
- Hybrid search
- Filters

Nice to Have

- Graph visualization
- Maintenance agent
- Compliance agent
- Lessons learned engine
- Notifications

10. Guiding Principles

Every engineering decision should satisfy these principles:

- One source of truth — never duplicate knowledge unnecessarily.
- AI must be grounded — every answer links back to source documents.
- Users search knowledge, not files — retrieval should be semantic, not just filename-based.
- Automation over manual work — new documents should automatically become part of the knowledge base.
- Modular by design — document processing, retrieval, AI, and storage should remain separate so they can evolve independently.